List of different interesting topics

The topics are designed to provide you with hands-on experience in cutting-edge machine/deep learning applications that are currently transforming industries worldwide. Each topic represents a real-world problem where AI is making a significant impact—from 3D printing to healthcare (brain tumor detection, skin cancer diagnosis. By working on these projects, you will not only strengthen your theoretical understanding of machine learning, neural networks, CNNs, transformers, and GANs, but also gain practical skills in implementing state-of-the-art models using industry-standard frameworks like PyTorch and TensorFlow.

Topic 1: Python-Based Automated Slicing and G-code Generation

Title: *Slice-100K: A Multimodal Dataset for Extrusion-based 3D Printing*

Dataset: Slice-100K Dataset — 100,000 STL/G-code pairs (NeurIPS 2024): https://github.com/idealab-isu/Slice-100K

Latest Papers: • Slice-100K: A Multimodal Dataset for Extrusion-based 3D Printing (NeurIPS 2024): https://arxiv.org/html/2407.04180v1

Topic 2: Real-Time 3D Printing Error Detection Using Python and Deep Learning

Title:
Generalisable 3D Printing Error Detection and Correction via Multi-Head Neural Networks

Dataset:
The Caxton dataset contains about 1.27 million labelled images of extrusion-based 3D printing. Images were captured near the printer nozzle across different parts, geometries, material colors, and lighting conditions. Python was used for sampling parameter combinations and cleaning/filtering the dataset.

Dataset URL: https://doi.org/10.17863/CAM.84082

Latest Papers:

- *Generalisable 3D printing error detection and correction via multi-head neural networks*: https://www.nature.com/articles/s41467-022-31985-y
- Dataset record: https://www.repository.cam.ac.uk/handle/1810/339869

Topic 3: FDM 3D Printing Defect Detection Using Python Computer Vision

Topic Title: Real-Time Defect Detection for Fused Filament Fabrication 3D Printing Using Lightweight Model Deployment

Dataset:

- *FDM 3D Printing Defect Dataset provides image data for detecting defects in fused deposition modeling / fused filament fabrication prints. It supports Python-based computer vision pipelines using OpenCV, PyTorch, TensorFlow, or YOLO models. Dataset URL: https://github.com/wengmhu/FDM-3D-Printing-Defect-Dataset*

Latest Paper:

- *Real-Time Defect Detection for Fused Filament Fabrication 3D Printing Using Lightweight Model Deployment: https://link.springer.com/article/10.1007/s00170-024-14452-4*
- *Quality Control in Additive Manufacturing: Image Dataset for Defect Detection in Fused Filament Fabrication:* https://zenodo.org/records/14712897

Topic 4: Python-Based Transfer Learning for 3D Printed Object Defect Detection

Title:
Transfer Learning-Based Defect Detection in 3D-Printed Objects

Dataset:
This topic uses image datasets of 3D-printed objects with defects. Python can be used with pretrained CNN models such as ResNet, VGG, MobileNet, EfficientNet, YOLOv5, and YOLOv8 for detecting defects in printed parts.

Dataset URL: https://github.com/mahsan2/AMdataset

GitHub project repository: https://github.com/mahsan2/AM3d

Latest Paper:

- Defect analysis of 3D printed object using transfer learning approaches: https://doi.org/10.1016/j.eswa.2024.124293

Topic 5: Python-Based STL Mesh Processing and In-Situ Repair

Title: *Enhancing 3D Printing with Procedural Generation and STL Formatting Using Python*

Dataset: PySLM Open-Source Python Library for 3D Printing and Additive Manufacturing (SLM/SLS/FDM)

https://github.com/drlukeparry/pyslm

https://cadquery.readthedocs.io/en/latest/intro.html

Latest Papers:

• Enhancing 3D Printing with Procedural Generation and STL Formatting Using Python: https://doi.org/10.3390/app14167299

Topic 6: Text-to-Image Generation

Topic Title: Generating Realistic Images from Text Descriptions using Diffusion Models

Dataset:

- LAION-5B - Large-scale Image-Text Dataset: https://laion.ai/blog/laion-5b/
- MS-COCO Captions: https://cocodataset.org/#captions-2015

Latest Paper:

- "SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis."
- Link: https://arxiv.org/abs/2307.01952
- Code: https://github.com/Stability-AI/generative-models

Topic 7: Human Pose Estimation

Topic Title: Multi-Person Human Pose Estimation using Deep Learning.

Dataset:

- COCO Key points: https://cocodataset.org/#keypoints-2020
- MPII Human Pose Dataset: http://human-pose.mpi-inf.mpg.de/
- Human3.6M: http://vision.imar.ro/human3.6m/

Latest Paper:

- "ViTPose: Simple Vision Transformer Baselines for Human Pose Estimation."
- Link: https://arxiv.org/abs/2204.12484
- Code: https://github.com/ViTAE-Transformer/ViTPose

Topic 8: Brain Tumor Segmentation Using Deep Learning

Topic Title: Automatic Brain Tumor Segmentation from MRI Images using U-Net and Transformer Models

Dataset:

- **BraTS 2023 - Brain Tumor Segmentation Challenge:** https://huggingface.co/datasets/AIOmarRehan/Brain_Tumor_MRI_Dataset
- **BraTS 2021:** https://www.kaggle.com/datasets/dschettler8845/brats-2021-task1

Latest Paper:

- "Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors."
- Link: https://arxiv.org/abs/2201.01266
- Code: https://github.com/Project-MONAI/research-contributions/tree/main/SwinUNETR